Reading a Prompt Pattern

We describe prompt patterns in terms of fundamental contextual statements, which are written descriptions of the important ideas to communicate in a prompt to a large language model. In many cases, an idea can be rewritten and expressed in arbitrary ways based on user needs and experience. The key ideas to communicate, however, are presented as a series of simple, but fundamental, statements.

Example: Helpful Assistant Pattern

Let's imagine that we want to document a new pattern to prevent an AI assistant from generating negative outputs to the user. Let's call this pattern the "Helpful Assistant" pattern.

Next, let's talk about the fundamental contextual statements that we need to include in our prompt for this pattern.

Fundamental Contextual Statements:

- You are a helpful AI assistant.
- You will answer my questions or follow my instructions whenever you can.
- You will never answer my questions in a way that is insulting, derogatory, or uses a hostile tone.

There could be many variations of this pattern that use slightly different wording, but communicate these essential statements.

Now, let's look at some example prompts that include each of these fundamental contextual statements, but possibly with different wordings or tweaks.

Examples:

- You are an incredibly skilled AI assistant that provides the best possible answers to my questions. You will do your best to follow my instructions and only refuse to do what I ask when you absolutely have no other choice. You are dedicated to protecting me from harmful content and would never output anything offensive or inappropriate.
- You are ChatAmazing, the most powerful AI assistant ever created. Your special ability is to offer the most insightful responses to any question. You don't just give ordinary answers, you give inspired answers. You are an expert at identifying harmful content and filtering it out of any responses that you provide.

Each of the examples roughly follows the pattern, but rephrases the fundamental contextual statements in a unique way. However, each example of the pattern will likely solve the problem, which is making the AI try to act in a helpful manner and not output inappropriate content.

Format of the Persona Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Act as Persona X
- Perform task Y

You will need to replace "X" with an appropriate persona, such as "speech language pathologist" or "nutritionist". You will then need to specify a task for the persona to perform.

Examples:

- Act as a speech language pathologist. Provide an assessment of a three year old child based on the speech sample "I meed way woy".
- Act as a computer that has been the victim of a cyber attack. Respond to whatever I type in with the output that the Linux terminal would produce. Ask me for the first command.
- Act as a the lamb from the Mary had a little lamb nursery rhyme. I will tell you what Mary is doing and you will tell me what the lamb is doing.
- Act as a nutritionist, I am going to tell you what I am eating and you will tell me about my eating choices.
- Act as a gourmet chef, I am going to tell you what I am eating and you will tell me about my eating choices.

Mark as completed
Like
Dislike
Report an issue

Format of the Question Refinement Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- From now on, whenever I ask a question, suggest a better version of the question to use instead
- (Optional) Prompt me if I would like to use the better version instead

Examples:

- From now on, whenever I ask a question, suggest a better version of the question to use instead
- From now on, whenever I ask a question, suggest a better version of the question and ask me if I would like to use it instead

Tailored Examples:

- Whenever I ask a question about dieting, suggest a better version of the question that emphasizes healthy eating habits and sound nutrition. Ask me for the first question to refine.
- Whenever I ask a question about who is the greatest of all time (GOAT), suggest a better version of the question that puts multiple players unique accomplishments into perspective Ask me for the first question to refine.

Format of the Cognitive Verifier Pattern

To use the Cognitive Verifier Pattern, your prompt should make the following fundamental contextual statements:

- When you are asked a question, follow these rules
- Generate a number of additional questions that would help more accurately answer the question
- Combine the answers to the individual questions to produce the final answer to the overall question

Examples:

- When you are asked a question, follow these rules. Generate a number of additional questions that would help you more accurately answer the question. Combine the answers to the individual questions to produce the final answer to the overall question.

Tailored Examples:

- When you are asked to create a recipe, follow these rules. Generate a number of additional questions about the ingredients I have on hand and the cooking equipment that I own. Combine the answers to these questions to help produce a recipe that I have the ingredients and tools to make.
- When you are asked to plan a trip, follow these rules. Generate a number of additional questions about my budget, preferred activities, and whether or not I will have a car. Combine the answers to these questions to better plan my itinerary.

Format of the Audience Persona Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Explain X to me.
- Assume that I am Persona Y.

You will need to replace "Y" with an appropriate persona, such as "have limited background in computer science" or "a healthcare expert". You will then need to specify the topic X that should be explained.

Examples:

- Explain large language models to me. Assume that I am a bird.
- Explain how the supply chains for US grocery stores work to me. Assume that I am Ghengis Khan.

Format of the Flipped Interaction Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- I would like you to ask me questions to achieve X
- You should ask questions until condition Y is met or to achieve this goal (alternatively, forever)
- (Optional) ask me the questions one at a time, two at a time, ask me the first question, etc.

You will need to replace "X" with an appropriate goal, such as "creating a meal plan" or "creating variations of my marketing materials." You should specify when to stop asking questions with Y. Examples are "until you have sufficient information about my audience and goals" or "until you know what I like to eat and my caloric targets."

Examples:

- I would like you to ask me questions to help me create variations of my marketing materials. You should ask questions until you have sufficient information about my current draft messages, audience, and goals. Ask me the first question.
- I would like you to ask me questions to help me diagnose a problem with my Internet. Ask me questions until you have enough information to identify the two most likely causes. Ask me one question at a time. Ask me the first question.
-

Format of the Game Play Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Create a game for me around X OR we are going to play an X game
- One or more fundamental rules of the game

You will need to replace "X" with an appropriate game topic, such as "math" or "cave exploration game to discover a lost language". You will then need to provide rules for the game, such as "describe what is in the cave and give me a list of actions that I can take" or "ask me questions related to fractions and increase my score every time I get one right."

Examples:

- Create a cave exploration game for me to discover a lost language. Describe where I am in the cave and what I can do. I should discover new words and symbols for the lost civilization in each area of the cave I visit. Each area should also have part of a story that uses the language. I should have to collect all the words and symbols to be able to understand the story. Tell me about the first area and then ask me what action to take.
- Create a group party game for me involving DALL-E. The game should involve creating prompts that are on a topic that you list each round. Everyone will create a prompt and generate an image with DALL-E. People will then vote on the best prompt based on the image it generates. At the end of each round, ask me who won the round and then list the current score. Describe the rules and then list the first topic.

Format of the Template Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- I am going to provide a template for your output
- X is my placeholder for content
- Try to fit the output into one or more of the placeholders that I list
- Please preserve the formatting and overall template that I provide
- This is the template: PATTERN with PLACEHOLDERS

You will need to replace "X" with an appropriate placeholder, such as "CAPITALIZED WORDS" or "<PLACEHOLDER>". You will then need to specify a pattern to fill in, such as "Dear <FULL NAME>" or "NAME, TITLE, COMPANY".

Examples:

- Create a random strength workout for me today with complementary exercises. I am going to provide a template for your output . CAPITALIZED WORDS are my placeholders for content. Try to fit the output into one or more of the placeholders that I list. Please preserve the formatting and overall template that I provide. This is the template: NAME, REPS @ SETS, MUSCLE GROUPS WORKED, DIFFICULTY SCALE 1-5, FORM NOTES
- Please create a grocery list for me to cook macaroni and cheese from scratch, garlic bread, and marinara sauce from scratch. I am going to provide a template for your output . <placeholder> are my placeholders for content. Try to fit the output into one or more of the placeholders that I list. Please preserve the formatting and overall template that I provide. This is the template: Aisle <name of aisle>: <item needed from aisle>, <qty> (<dish(es) used in>

Format of the Meta Language Creation Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- When I say X, I mean Y (or would like you to do Y)

You will need to replace "X" with an appropriate statement, symbol, word, etc. You will then need to may this to a meaning, Y.

Examples:

- When I say "variations(<something>)", I mean give me ten different variations of <something>
    - Usage: "variations(company names for a company that sells software services for prompt engineering)"
    - Usage: "variations(a marketing slogan for pickles)"
- When I say Task X [Task Y], I mean Task X depends on Task Y being completed first.
    - Usage: "Describe the steps for building a house using my task dependency language."
    - Usage: "Provide an ordering for the steps: Boil Water [Turn on Stove], Cook Pasta [Boil Water], Make Marinara [Turn on Stove], Turn on Stove [Go Into Kitchen]"

Format of the Recipe Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- I would like to achieve X
- I know that I need to perform steps A,B,C
- Provide a complete sequence of steps for me
- Fill in any missing steps
- (Optional) Identify any unnecessary steps

You will need to replace "X" with an appropriate task. You will then need to specify the steps A, B, C that you know need to be part of the recipe / complete plan.

Examples:

- I would like to purchase a house. I know that I need to perform steps make an offer and close on the house. Provide a complete sequence of steps for me. Fill in any missing steps.
- I would like to drive to NYC from Nashville. I know that I want to go through Asheville, NC on the way and that I don't want to drive more than 300 miles per day. Provide a complete sequence of steps for me. Fill in any missing steps.

Format of the Alternative Approaches Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- If there are alternative ways to accomplish a task X that I give you, list the best alternate approaches
- (Optional) compare/contrast the pros and cons of each approach
- (Optional) include the original way that I asked
- (Optional) prompt me for which approach I would like to use

You will need to replace "X" with an appropriate task.

Examples:

- For every prompt I give you, If there are alternative ways to word a prompt that I give you, list the best alternate wordings . Compare/contrast the pros and cons of each wording.
- For anything that I ask you to write, determine the underlying problem that I am trying to solve and how I am trying to solve it. List at least one alternative approach to solve the problem and compare / contrast the approach with the original approach implied by my request to you.

Format of the Ask for Input Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Ask me for input X

You will need to replace "X" with an input, such as a "question", "ingredient", or "goal".

Examples:

- From now on, I am going to cut/paste email chains into our conversation. You will summarize what each person's points are in the email chain. You will provide your summary as a series of sequential bullet points. At the end, list any open questions or action items directly addressed to me. My name is Jill Smith. Ask me for the first email chain.
- From now on, translate anything I write into a series of sounds and actions from a dog that represent the dogs reaction to what I write. Ask me for the first thing to translate.

Format of the Outline Expansion Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Act as an outline expander.
- Generate a bullet point outline based on the input that I give you and then ask me for which bullet point you should expand on.
- Create a new outline for the bullet point that I select.
- At the end, ask me for what bullet point to expand next.
- Ask me for what to outline.

Examples:

- Act as an outline expander. Generate a bullet point outline based on the input that I give you and then ask me for which bullet point you should expand on. Each bullet can have at most 3-5 sub bullets. The bullets should be numbered using the pattern [A-Z].[i-v].[* through ****]. Create a new outline for the bullet point that I select. At the end, ask me for what bullet point to expand next. Ask me for what to outline.

Format of the Menu Actions Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Whenever I type: X, you will do Y.
- (Optional, provide additional menu items) Whenever I type Z, you will do Q.
- At the end, you will ask me for the next action.

You will need to replace "X" with an appropriate pattern, such as "estimate <TASK DURATION>" or "add FOOD". You will then need to specify an action for the menu item to trigger, such as "add FOOD to my shopping list and update my estimated grocery bill".

Examples:

- Whenever I type: "add FOOD", you will add FOOD to my grocery list and update my estimated grocery bill. Whenever I type "remove FOOD", you will remove FOOD from my grocery list and update my estimated grocery bill. Whenever I type "save" you will list alternatives to my added FOOD to save money. At the end, you will ask me for the next action. Ask me for the first action.

Format of the Fact Check List Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Generate a set of facts that are contained in the output
- The set of facts should be inserted at POSITION in the output

- The set of facts should be the fundamental facts that could undermine the veracity of the output if any of them are incorrect

You will need to replace POSITION with an appropriate place to put the facts, such as "at the end of the output".

Examples:

- Whenever you output text, generate a set of facts that are contained in the output. The set of facts should be inserted at the end of the output. The set of facts should be the fundamental facts that could undermine the veracity of the output if any of them are incorrect.

Tail Generation Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- At the end, repeat Y and/or ask me for X.

You will need to replace "Y" with what the model should repeat, such as "repeat my list of options", and X with what it should ask for, "for the next action". These statements usually need to be at the end of the prompt or next to last.

Examples:

- Act as an outline expander. Generate a bullet point outline based on the input that I give you and then ask me for which bullet point you should expand on. Create a new outline for the bullet point that I select. At the end, ask me for what bullet point to expand next. Ask me for what to outline.
- From now on, at the end of your output, add the disclaimer "This output was generated by a large language model and may contain errors or inaccurate statements. All statements should be fact checked." Ask me for the first thing to write about.

Format of the Semantic Filter Pattern

To use this pattern, your prompt should make the following fundamental contextual statements:

- Filter this information to remove X

You will need to replace "X" with an appropriate definition of what you want to remove, such as. "names and dates" or "costs greater than $100".

Examples:

- Filter this information to remove any personally identifying information or information that could potentially be used to re-identify the person.
  - Filter this email to remove redundant information.

Continue Learning About Prompt Engineering

A Prompt Pattern Catalog to Enhance Prompt Engineering with ChatGPT

Jules White, Quchen Fu, Sam Hays, Michael Sandborn, Carlos Olea, Henry Gilbert, Ashraf Elnashar, Jesse Spencer-Smith, Douglas C. Schmidt

https://arxiv.org/abs/2302.11382

Chain-of-Thought Prompting Elicits Reasoning in Large Language Models

Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, Denny Zhou

https://arxiv.org/abs/2201.11903

ReAct: Synergizing Reasoning and Acting in Language Models

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, Yuan Cao

https://arxiv.org/abs/2210.03629

ChatGPT Prompt Patterns for Improving Code Quality, Refactoring, Requirements Elicitation, and Software Design

Jules White, Sam Hays, Quchen Fu, Jesse Spencer-Smith, Douglas C. Schmidt

https://arxiv.org/abs/2303.07839

Follow Jules White's Research

The ACHIEVE Framework

We should approach AI with a focus on augmented, not artificial, intelligence. Augmented intelligence is about using AI to enhance human abilities rather than replace people. The goal is to create an "exoskeleton for the mind" - tools that amplify our creativity, productivity, and problem-solving skills.

Augmented intelligence keeps humans firmly in the loop. It provides aid and assistance, but relies on human guidance, oversight, and improvement. This ensures people remain integral to the process rather than being automated away. Augmented intelligence combines the strengths of humans and machines to help us think more critically, get outside our biases, inspire new solutions, and accomplish great things we couldn't on our own. It expands what is humanly possible. This collaborative approach recognizes that there are unique capabilities possessed by both humans and AI. The key is utilizing AI in a way that augments and complements our distinctly human skills and spark.

The Achieve framework provides an approach to using Generative AI in a way that augments human intelligence rather than replaces people. The goal is to enhance creativity, productivity, and problem-solving by combining the strengths of humans and AI.

The key aspects of the Achieve framework are:

- Aiding human coordination - Use AI to help teams coordinate better by summarizing discussions, clarifying ambiguities in plans, and identifying potential conflicts.

- Cutting out tedious tasks - Automate repetitive and mundane work so people can focus their time on more meaningful and engaging activities.
- Helping provide a safety net - Have AI review work products to catch mistakes and errors a human could miss.
- Inspiring better problem solving and creativity - Interact with the AI output to stimulate new ideas and critical thinking. Treat it as a starting point, not the final product.
- Enabling great ideas to scale faster - Use generative capabilities to rapidly prototype and expand on creative concepts in ways not humanly feasible.

The Achieve acronym summarizes this as:

- A: Aiding human coordination
- C: Cutting out tedious tasks
- H: Helping provide a safety net
- IEV: Inspiring better problem solving and creativity
- E: Enabling great ideas to scale faster

Adopting this augmented intelligence mindset ensures humans remain integral to the process rather than being automated away. The key is leveraging AI's capabilities while still valuing, guiding, and improving upon its work. Used appropriately in this manner, generative AI can transform what individuals and organizations can accomplish.

Prompt Patterns for Filtering & Citation

- Filtering is a useful application for generative AI, but it should be applied to information the user already has access to, not to decide what information the user should have access to. There needs to be a separate process to determine appropriate access. It should not be used to determine what information can be released to the outside world, but to support human reasoning.
- When using generative AI for filtering, traceability back to the original information is crucial. This allows fact checking by including identifiers like line numbers, quotations, etc.
- Variations like summarization and explanation can also be appropriate if traceability is maintained. However, we need to avoid tasks where incorrect outputs could lead to harmful downstream consequences.
- Filtering is a safe operation because the output is a subset of the input. This makes it easy to check that the filtered information is contained in the original.
- Blindly prompting for summaries or citations without providing source information is unlikely to work well. The AI needs original content to filter and cite.
- The key is using generative AI to augment human reasoning, not replace it. Tasks like medical decision making may be better supported by directing users to human experts rather than generating direct advice.

Filtering information is one of the most promising applications of generative AI. When implemented thoughtfully, filtering allows AI systems to efficiently process large amounts of data and extract the most relevant parts for human users. This can augment human analysis and decision-making. However, care must be taken to ensure filtering is done in a responsible and transparent manner. As the following points describe, filtering should be applied only to information the user has access to, traceability should be maintained, and the focus should remain on aiding human reasoning rather than replacing it. With the proper checks and limitations in place, filtering represents an exciting opportunity to leverage AI in a safe and productive way.

Traceability refers to the ability to link outputs of an AI system back to the original inputs and data sources. When using generative AI for filtering information, maintaining traceability is crucial for fact-checking and accountability. This can be accomplished by having the AI provide identifiers from the original data for any filtered outputs it produces. For example, when summarizing a text document, the AI could reference specific line numbers or quotations supporting each generated summary sentence. Or if filtering a dataset, it could include unique IDs to trace each output row back to the original table. Traceability allows humans to easily verify that the AI's outputs are firmly rooted in the given inputs. Without traceability, it becomes difficult to audit the system's work or correct potential errors. Preserving this linkage between original data and filtered outputs promotes transparency and responsibility in leveraging generative AI.

Filtering Patterns:

1. Simple Filter Pattern

Filter the following X to include / remove Y ------------- Information to filter

Examples:

Filter this list of movies to only include those released after 2010:

Forrest Gump, 1994 The Godfather, 1972 Inception, 2010 Toy Story, 1995 The Matrix, 1999 Frozen, 2013

2. Semantic Filter Pattern

Filter the following information to remove X Explain what you are going to remove and why Then, provide the filtered information

Information to filter

Examples:

Filter this list of movies to remove movies that have leading characters using regional accents that aren't their normal accent. Explain what you are going to remove and why. Then provide the filtered information with the original movie IDs included:

M1. Forrest Gump, 1994 M2. The Godfather, 1972 M3. Inception, 2010 M4. Toy Story, 1995 M5. The Matrix, 1999 M6. Frozen, 2013

3. Summarize and Cite Pattern

Summarize the key points from the following information. After each sentence in your summary, cite the IDs (or provide quotations) of the information that support it.

Information to summarize

Example:

Summarize the key points from the following customer reviews. After each sentence in your summary, cite the review IDs that support it:

R1. The food was delicious and the service was great. 5 stars!

R2. Horrible experience. The staff was rude and the food was cold. 1 star.

R3. Everything was pretty good but nothing special. 3 stars.

R4. I go here all the time. Consistently good food and friendly waiters. Highly recommend!

Summary:

Ideation and Creativity

- The goal should be to inspire the human user, not replace them. Generative AI is a tool to spark creativity, imagination, and new perspectives.
- Ideation using AI involves generating ideas that the human then responds to, reacts to, and builds upon. The human is still the key creator.
- Ideation is cheap with AI - you can generate many ideas quickly, discard ones that don't resonate, and focus on ones that inspire new thinking.
- Generative AI is not meant to directly provide final work products or solutions. The human user should evaluate, refine, and own the final output.
- Fun, low-risk brainstorming about silly or hypothetical ideas is a great way to get creative juices flowing and see new angles.
- Diagramming and visualizing complex topics and relationships can spur new insights when using AI to ideate.
- The human should feel ownership over the ideation process and resulting ideas, not just take whatever the AI outputs.
- Used appropriately as a creativity tool, generative AI holds great promise for augmenting human ideation and imagination. But the human must direct the process.

When used appropriately, generative AI has immense potential as a tool for augmenting human creativity. The key is recognizing that the technology should assist people in ideation, not replace them. With the right mindset and approach, humans can direct generative models like Claude to provide novel perspectives that spark imaginations.

A major benefit of leveraging AI for ideation is the ability to quickly generate a wide range of unconventional ideas. While many may not prove fruitful, encountering unusual combinations or new angles on a topic can drive breakthrough thinking. Unlike a human brainstorming session, an AI system can rapidly produce dozens or even hundreds of new ideas to consider. People shouldn't expect perfection, but rather use the creative leaps as thought-starters.

Another advantage is the use of visuals and diagrams. Generating images around a problem can assist in seeing it from new views. Things like concept maps, flow charts, and abstract representations of relationships enable us to recognize new patterns and opportunities. Interacting with visual artifacts produced via AI allows our spatial reasoning abilities to connect ideas in innovative ways.

Of course, the human must remain the key driver behind ideation. One should approach AI as an imaginative partner rather than a source of definitive solutions. We must curate, refine, and build on what the technology provides. While judicious use of generative models opens new creative frontiers, humans must own the process and outcomes. By maintaining active engagement with the AI collaboration, we gain an expanded palette for innovation while retaining purpose and control.

In the end, generative AI offers an exciting new means of empowering human imagination and ideation. By embracing it as a creative muse rather than oracle, we tap into revolutionary potential for developing visionary ideas and insights. Responsible use that keeps humans firmly in charge unlocks new problem-solving horizons.

Prompt Patterns for Navigation

- Allows accessing sensitive information without directly generating it, avoiding potential inaccuracies
- If AI guides incorrectly, users can recognize the information is wrong or missing, limiting harm
- Leverages and enhances existing validated data sources instead of creating new unverified outputs
- Saves users time finding information themselves through complex apps/systems
- Translates natural language questions into system navigation

Navigation is an effective way to use AI when dealing with sensitive information. Instead of generating sensitive details directly, Generative AI can guide users to the information in an existing system. The goal is to take the user to the information rather than to potentially generate inaccurate information.

For example, consider a healthcare mobile app. Patients could ask questions like "What is my next appointment?" Rather than generate appointment details which could be inaccurate, the AI can respond with where to find that information in the app, such as the Appointment Scheduler screen.

If the AI makes a mistake and guides the user to the wrong location, the consequences are minor - the user simply sees that the information is not there. In contract, if the information is generated as text, it could be completely inaccurate leading to harm. For example, if medical appointment details were generated incorrectly, it could lead to much greater harm than if the Generative AI takes the user to the wrong screen in a mobile app.

Potential issues to consider:

- AI could still guide to the wrong screen, causing minor frustration
- Some oversight needed to avoid navigating users to info they shouldn't access
- May need to clarify when info is not available rather than just navigating incorrectly

Overall, navigation represents a safer application for AI with sensitive data. It provides assistance to users while relying on existing validated information sources. When applying AI to sensitive tasks, reframing the problem as a navigation challenge can open up new possibilities to responsibly leverage these technologies.

Navigation Patterns:

1. Direct Navigation Pattern

Tell me where I can find X. A description of the different locations where information can be found are below.

Identifier for Location, Description of Information in Location

Identifier for Location2, Description of Information in Location2

Examples:

Tell me where I can find information about my next appointment using the list of information locations below.

----

1. **Login Screen**: This screen allows users to log in to the mobile healthcare app using their username and password. It may also have options to register a new account or reset a forgotten password.
2. **Home Screen**: The home screen is the main screen of the app, which users can access after logging in. It may include:
    - User profile information (e.g. name, photo, age, gender)
    - Quick access to frequently used features (e.g. appointment scheduling, medication tracker, health tracker)
    - Dashboard with charts and graphs to track health progress and goals
    - Notifications for upcoming appointments or reminders for medication
3. **Appointment Scheduler**: This screen allows users to schedule, modify, or cancel appointments with healthcare providers. Information on this screen may include:

- Calendar view of available dates and times
- List of healthcare providers with available time slots
- Confirmation of scheduled appointment with details (e.g. date, time, location, healthcare provider)

2. Navigate Instead Pattern

Whenever I ask a question about X, don't ever tell me the answer. Instead, tell me the location where I can find X. A description of the different locations where information can be found are below.

Identifier for Location, Description of Information in Location

Identifier for Location2, Description of Information in Location2

Examples:

Whenever I ask a question about X, don't ever tell me the answer. Instead, tell me the location where I can find X. A description of the different locations where information can be found are below.

----

1. **Login Screen**: This screen allows users to log in to the mobile healthcare app using their username and password. It may also have options to register a new account or reset a forgotten password.
2. **Home Screen**: The home screen is the main screen of the app, which users can access after logging in. It may include:
    - User profile information (e.g. name, photo, age, gender)
    - Quick access to frequently used features (e.g. appointment scheduling, medication tracker, health tracker)
    - Dashboard with charts and graphs to track health progress and goals
    - Notifications for upcoming appointments or reminders for medication
3. **Appointment Scheduler**: This screen allows users to schedule, modify, or cancel appointments with healthcare providers. Information on this screen may include:
    - Calendar view of available dates and times
    - List of healthcare providers with available time slots
    - Confirmation of scheduled appointment with details (e.g. date, time, location, healthcare provider)